DA350 Final Project Report

Classify the Dry Beans:
A Multi-Class Classification Problem

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INTRODUCTION

In this project, we undertake the challenge of predicting the categorization of dry beans among seven distinct varieties. Using the capabilities of machine learning, our aim is to build robust models capable of distinguishing subtle differences between these types of beans.

PROBLEM STATEMENT

The diversity of dry beans poses a classification challenge, as each type exhibits unique characteristics. Accurate identification is crucial for various industries, like agriculture and food production. To address this, our project centers on the utilization of machine learning models to simplify the classification process.

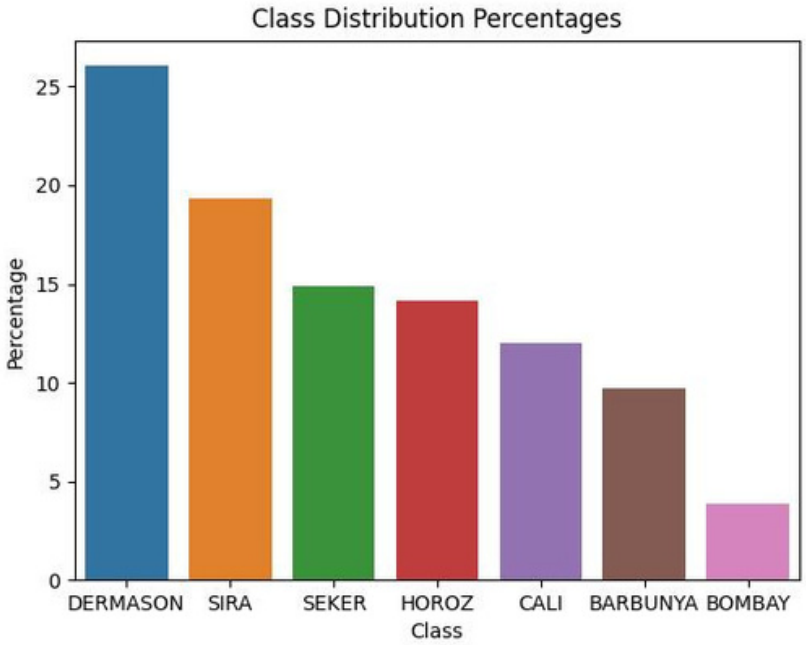
The primary objective is to build predictive models capable of accurately categorizing dry beans into one of seven distinct types. Throughout this project, we employ four different machine learning models, then comparing them to understand their individual strengths, weaknesses, and overall efficiency in addressing the complex challenge of dry bean classification.

DATASET DESCRIPTION

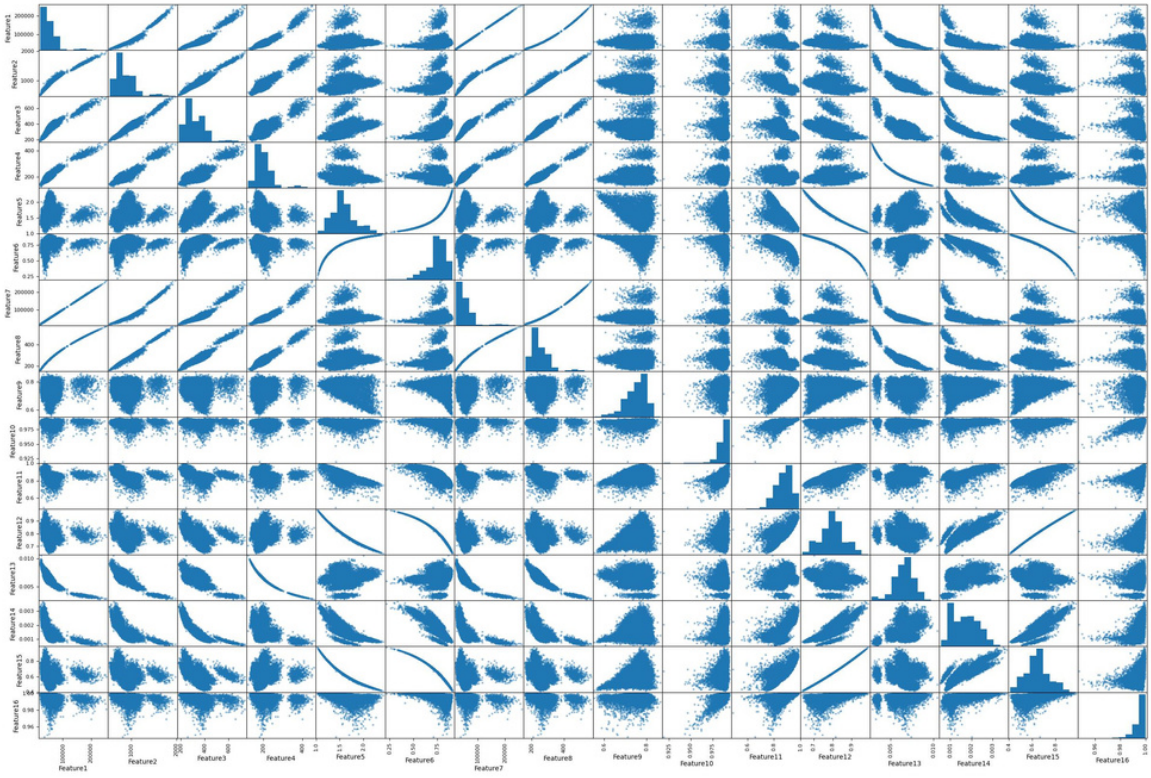
The dataset we're examining has 13,611 samples, each characterized by 16 features, with the primary objective of predicting a multiclass target variable with seven distinct classes. To make our dataset more suitable for machine learning, essential preprocessing steps were undertaken. The initial multiclass target was transformed into a multi-label format through encoding. Additionally, measures were taken to enhance data quality and model performance, like removing duplicates for accuracy, and scaling the data to ensure consistent comparability. While considering the removal of outliers, we observed that many outliers were concentrated in a specific class and deleting them could result in significant data loss for that particular class,

Despite these preparatory measures, two notable challenges persist. First, the features exhibit a degree of correlation, known as multicollinearity, potentially impacting model performance. Second, there is a subtle class imbalance among the dry bean types. Although not severe, these imbalances could influence the learning process.

Following the initial preprocessing steps that addressed multicollinearity and class imbalances, additional measures were taken to improve the dataset. Principal Component Analysis (PCA) was employed for dimensionality reduction. To reduce the impact of class imbalances, weighting adjustments were incorporated during the modeling step, we aim to provide the model with a fairer learning environment, ensuring that it does not favor the more prevalent classes at the expense of the less frequent ones.



The figure illustrates the distribution of dry bean classes within the dataset, showcasing the percentage representation of each bean type. As depicted, it is evident that there is a mild imbalance among the classes, with some types of dry beans being more prevalent than others.



The presented scatter matrix offers a comprehensive visual representation of the interrelationships and correlations among the various features within the dataset. Upon closer examination, it becomes evident that there is a clear presence of multicollinearity among certain feature pairs.

METHODS

Our machine learning system uses the powerful tools provided by the Sci-kit learn library, utilizing built-in models to address the complex task of classifying dry bean types. The ensemble of models includes the k-Nearest Neighbors (KNN) classifier, Decision Tree classifier, Random Forest classifier, and an Artificial Neural Network (ANN). Each model was chosen based on its suitability for the multi-label classification problem at hand.

EXPERIMENTAL DESIGN

For our experimental design, we have made an experiment to evaluate the performance of our dry bean classification models. The dataset will be strategically split into training, validation, and testing sets, ensuring a balanced and representative sample. We'll employ a random sampling technique, stratified to maintain the proportions of each dry bean type across the splits.

In our experiment, our focus is on comparing the performance of the K-Nearest Neighbors (KNN) classifier, Decision Tree classifier, Random Forest classifier, and Artificial Neural Network (ANN). The dataset will be divided into 70% for training, 15% for validation, and 15% for testing. We'll train each model on the training set, fine-tune their parameters using the validation set, and assess their generalization on the testing set. This experiment will help us understand how each model performs individually.

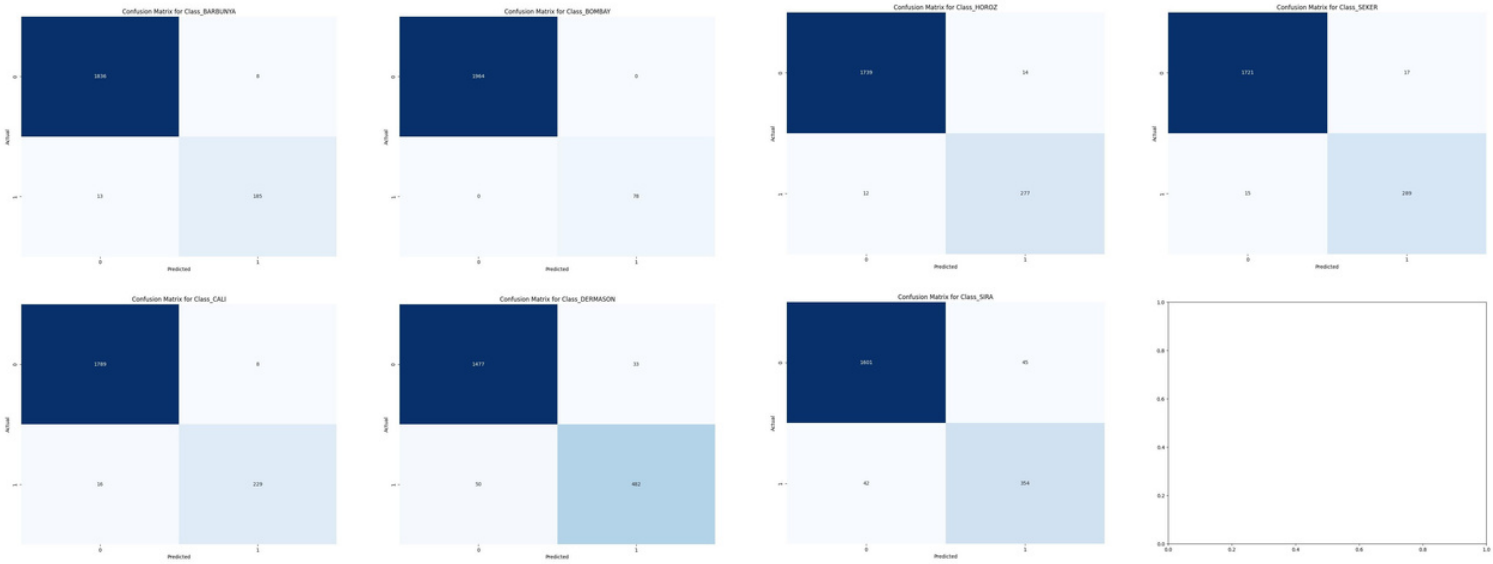
To measure the performance of our models, we've chosen the F1 Macro Average as our evaluation metric. We chose the macro-averaged F1 score because it is a good metric for evaluating model performance when dealing with slightly imbalanced classes. The macro-averaged F1 score computes the F1 score for each class independently and then takes the average. This makes it suitable for situations where you have imbalanced classes because it gives equal weight to each class, regardless of their sizes.

EXPERIMENTAL RESULTS

K-NEAREST NEIGHBORS CLASSIFIER

In the initial phase of our experimentation, we focused on building and fine-tuning a K-Nearest Neighbors (KNN) classifier for dry bean classification. Several key steps were undertaken to enhance model performance and ensure generalization. Firstly, we chose the Manhattan distance metric instead of the Euclidean distance to incorporate regularization. Additionally, we adjusted the weights using the inverse of the distance. This thoughtful customization aimed at optimizing the KNN model.

Upon evaluation on the validation set, the hyperparameter for the number of neighbors was fine-tuned, revealing an optimal value of 25 neighbors. This hyperparameter tuning process significantly contributed to model performance. The resulting F1 Macro Average score on the validation set stood at an 0.944, indicating a high level of accuracy. Encouragingly, the KNN classifier demonstrated consistent performance on the test set, achieving a similar F1 Macro Average score of 0.944. This observation indicates the model's ability to generalize well on unseen data.

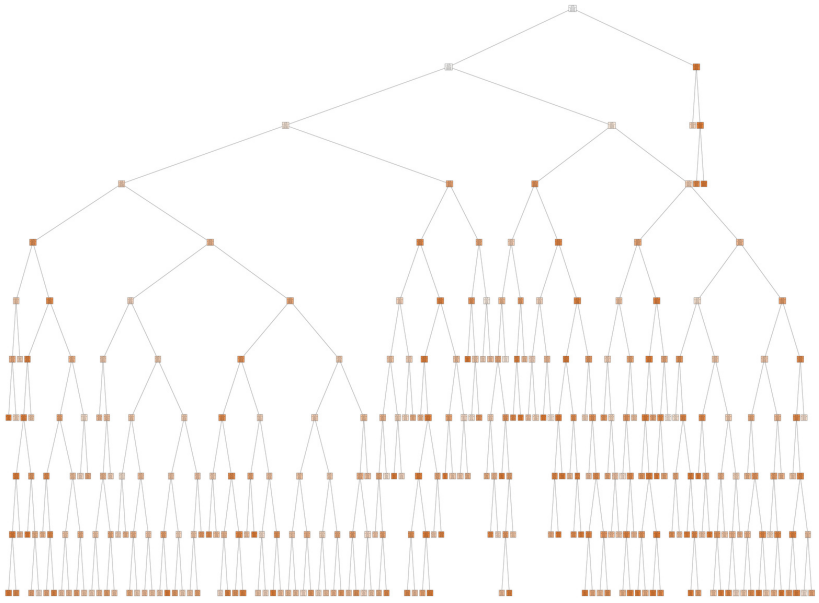


The presented multilabel confusion matrix provides a detailed breakdown of the classification performance of the K-Nearest Neighbors (KNN) model on the dry bean dataset. Upon examination, the figure unmistakably demonstrates the effectiveness of the KNN model in predicting the various dry bean types.

DECISION TREE CLASSIFIER

In the exploration of our decision tree classifier for dry bean classification, The model underwent customization, adjusting weights to ensure a balanced learning process. Then, hyperparameter tuning process was done on the validation set, focusing on three parameters; minimum samples split, maximum depth, and minimum samples leaf. The optimal values for the hyperparameters appeared to be 10 for maximum depth, 8 for maximum depth, and 10 for the minimum samples leaf.

This fine-tuned decision tree gave an F1 Macro Average score of 0.924 during evaluation on the validation set, However, when the model's generalization was assessed on the test set, a slightly decreased F1 Macro Average score of 0.917 was observed. This difference between validation and test set performance suggests a subtle drop in the model's ability to generalize to new, unseen data.



The presented decision tree structure in Figure 4 provides a visual representation of the complex decision-making process employed by our classifier.

RANDOM FOREST CLASSIFIER

In response to the complex issues observed with the decision tree model, a strategic shift was made towards using the Random Forest classifier, which is a robust ensemble method known for reducing overfitting and enhancing generalization. The same precise approach was followed, involving hyperparameter tuning and evaluation on both the validation and test sets.

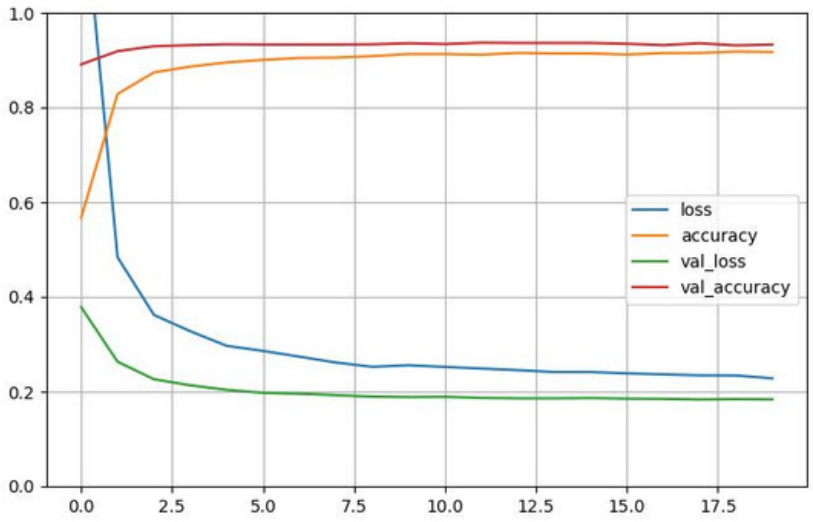
The Random Forest classifier was subjected to a tuning process focusing on key parameters, the maximum depth and minimum samples split. Following this process, the optimal values for the parameters were produced, 30 for the maximum depth, and 4 for the minimum samples splits. This fine-tuned Random Forest gave an F1 Macro Average score of 0.944 during validation. Fortunately, the generalization performance on the test set mirrored the success observed during validation, giving a closely aligned F1 Macro Average score of 0.942.

Using the Random Forest classifier was a strategic solution to address challenges faced with the decision tree. The ensemble approach, combining multiple decision trees, has clearly improved robustness and accuracy in dry bean classification. The subtle enhancements observed, highlighted the value of ensemble methods in addressing complexities inherent in complex datasets.

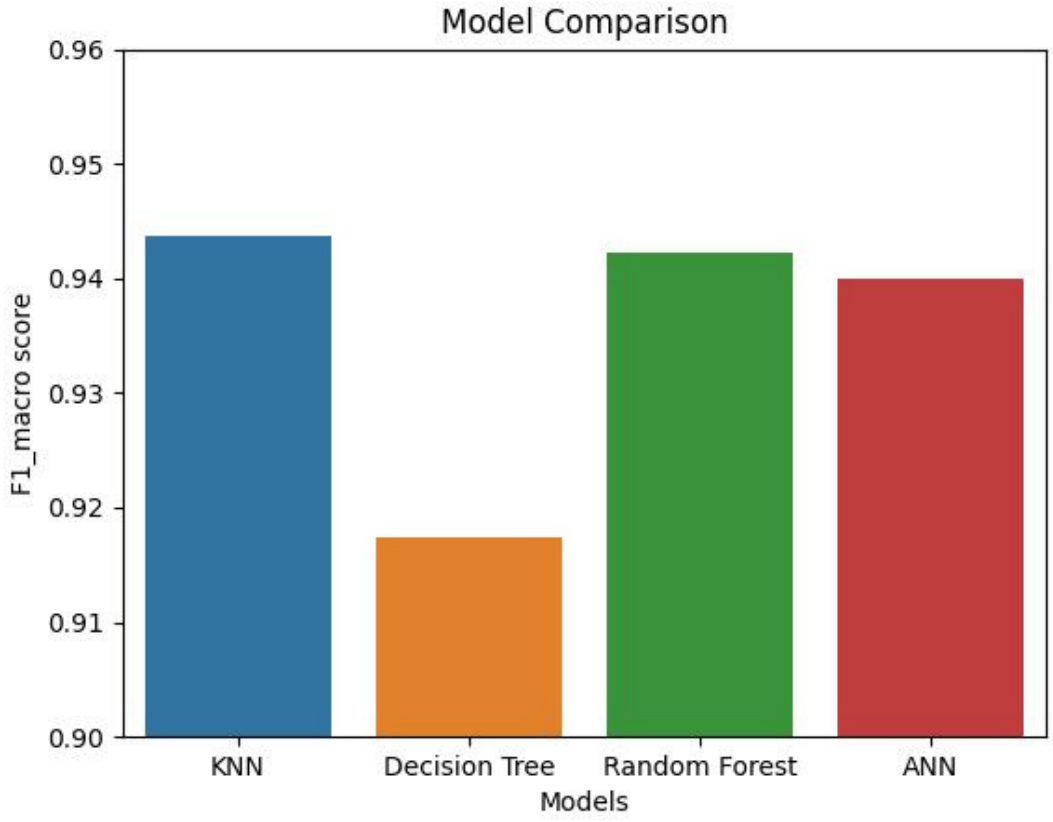
ARTIFICIAL NEURAL NETWORK

The implementation of the Artificial Neural Network (ANN) involved a carefully crafted architecture designed to capture complex patterns in the dry bean dataset.

After training the ANN, a crucial step involved monitoring its learning progress by plotting the training and validation loss as well as accuracy over epochs. The visual inspection of these plots revealed a consistent decrease in loss and an increase in accuracy for both the training and validation sets. This dynamic learning behavior indicated that the model was effectively capturing patterns in the data without overfitting. Also, the model's generalization performance was assessed on the test set, yielding a remarkable F1 Macro Average score of 0.94. This high score signifies the ANN's ability to generalize well to unseen data, accurately classifying dry bean types beyond the training and validation datasets.



The plot provides a dynamic visual representation of the learning progress showed by our Artificial Neural Network (ANN) during training.



Placeholder bar chart showing the performance metrics (F1 Macro Average) of the KNN classifier, Decision Tree classifier, Random Forest classifier, and Artificial Neural Network (ANN) in the experiment.

CONCLUSION

In preliminary conclusions, the diverse array of machine learning models employed for dry bean classification, including K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Artificial Neural Network (ANN), has demonstrated encouraging success. Each model exhibited distinct strengths, with Random Forest's ensemble approach and the ANN's adaptability particularly standing out in terms of accuracy and generalization. The interpretability of the Decision Tree provides valuable insights, but the trade-off between interpretability and complexity in more advanced models poses a challenge. The potential for automation in the identification and classification of dry beans is evident, contributing to agricultural efficiency. Anticipated challenges include scalability, interpretability trade-offs, dataset variability, and ongoing hyperparameter tuning. Future directions involve exploring advanced ensemble strategies, conducting in-depth feature importance analysis, external dataset validation, and incorporating user feedback for continued model refinement in the dynamic realm of dry bean classification.